

AIM Serial I/F Design Specification

Ver 0.1

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver 0.1	04/15/02	Hisanori Terawa	Initial Release for DM270.

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1. ABSTRACT

ARM Internal Memory (AIM) can be accessed only by ARM. AIM Serial I/F (AIMSIF) make access from out of the device with the serial access.

2. REFERENCES

ARM internal memory design specification

3. GLOSSARY

AIM	ARM Internal Memory
AIMSIF	AIM Serial Interface
ARM	ARM71TDMIE

4. FUNCTION

AIMSIF is given the three signals serial clock, enable and data in and write to AIM. In addition, AIMSIF can read data from AIM and output to data out serial pin.

The design target for AIMSIF module is as follows:

- ◆ Full addressing from 0x0 to 0x007F:FFFF can be done.
- ◆ Access mode is a byte mode only.
- ◆ Address can be incremented automatically after gotten first address.
- ◆ Support both of write and read mode.
- ◆ Effectively during global reset.
- ◆ senz=1の時はsclkを入力しない。

4.1 Block Diagram

The following shows the block diagram of AIMSIF.

Figure 1 : Block Diagram of AIMSIF

4.2 Functionality

The first bit received is the command bit and the address bits are 15 bits following the command bit. When the command bit is low, being the write mode and when the command bit is high, being the read mode.

Following the address, data of 8 bits length can be received in the write mode.

In the read mode, data of 8 bits length read from AIM will be transferred through data out pin.

The enable signal is active low. The address to be accessed is incremented automatically during enable signal is low.

The minimum unit of the transfer is defined 8 bits. If the data are less than 8 bits will be discarded at the last transferring unit.

The data to be transferred is MSB first.

The period of clock is 10MHz.

AIMSIF is performed while the global reset signal is in assertion.

4.3 Timings

The following timing chart shows the write mode and the read mode.

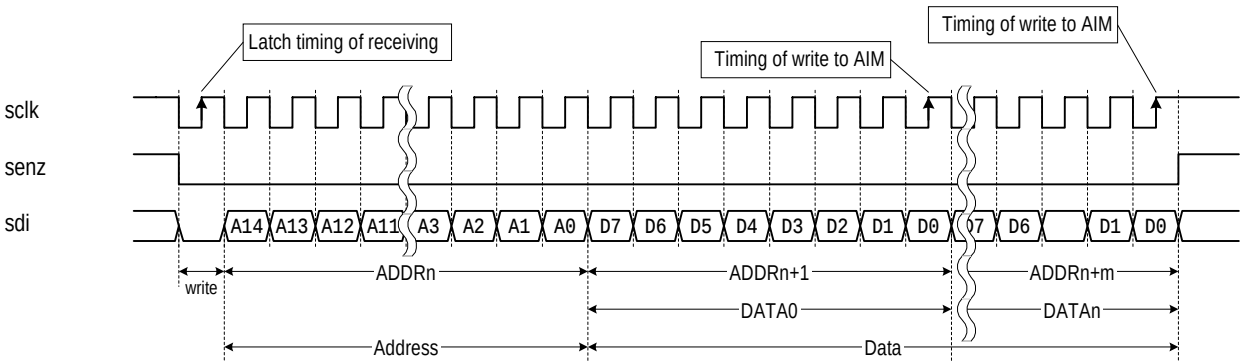


Figure 2 : Write mode (receive)

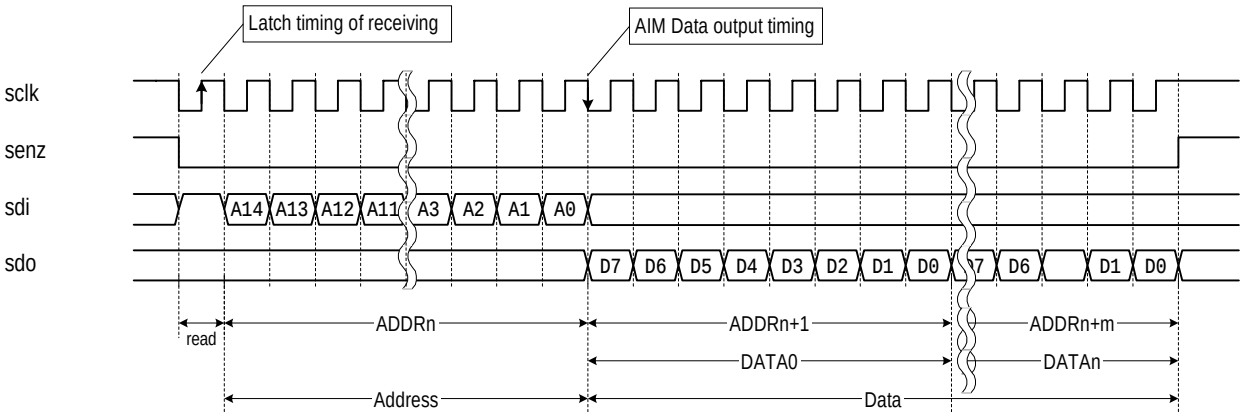


Figure 3 : Read mode (send)

5. REGISTERS

AIMSIF has no register.

6. Interface

Table 1 : AIMSIF Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Common signals				
rst_n	1	I	ext	Power on reset..
Serial interface signals				
sclk	1	I	ext	Serial clock.
nsclk	1	I		Serial clock negative.
senz	1	I		Serial enable signal. Active low.
sdi	1	I		Serial data input.
aimsif_sdo	1	O		Serial data output.
ARM Internal memory interface signals				
aimsif_ma	11	O	AIM	Internal memory address bus.
aimsif_mdo	32	O	AIM	Internal memory data output bus.
aim_do_aimsif	32	I	AIM	Internal memory data input bus.
aimsif_mrnw	1	O	AIM	Internal memory 1:Read / 0:Write.
aimsif_ncs_aim	4	O	AIM	Internal memory chip select. Active low.
aimsif_nds	4	O	AIM	Internal memory chip select. Active low.

7. TEST Methodology

none.